Cell Biology — Chapter 3: Cell Structure and Functio

3.1 Introduction to Cell Structure

The cell is defined as the smallest structural and functional unit of all living organisms. Every organism, whether unicellular such as bacteria or multicellular such as humans, is composed of one or more cells. Cell theory states that all living things are composed of cells, the cell is the basic unit of life, and all cells arise from pre-existing cells.

A eukaryotic cell is a cell that contains a true, membrane-bound nucleus along with other membrane-bound organelles. In contrast, a prokaryotic cell is a cell that lacks a nucleus and membrane-bound organelles; bacteria and archaea are examples of prokaryotic organisms. Understanding the differences between eukaryotic and prokaryotic cells is essential for understanding how life is organized at the microscopic level.

The plasma membrane, also called the cell membrane, is defined as a selectively permeable barrier composed of a phospholipid bilayer that separates the internal contents of the cell from the external environment. It regulates what substances enter and exit the cell through processes such as diffusion, osmosis, and active transport.

3.2 The Nucleus and Genetic Material

The nucleus is defined as the membrane-bound organelle that houses the cell's genetic material in the form of chromatin, which condenses into chromosomes during cell division. The nucleus is often referred to as the control center of the cell because it directs protein synthesis and cellular activity by regulating gene expression.

DNA replication refers to the biological process by which a cell duplicates its DNA before cell division, ensuring that each daughter cell receives an identical copy of the genetic material. The process relies on the enzyme DNA polymerase, which reads the existing strand as a template and synthesizes a new complementary strand.

The nucleolus is a dense region within the nucleus that is responsible for the production of ribosomal RNA and the assembly of ribosomal subunits. Although it is not surrounded by its own membrane, the nucleolus is a distinct and essential structure within the nucleus.

3.3 Organelles and Their Functions

Mitochondria are membrane-bound organelles that generate most of the cell's supply of adenosine triphosphate, commonly abbreviated as ATP, through a process called cellular respiration. Because of this role, mitochondria are frequently called the powerhouse of the cell. The equation for cellular respiration can be summarized as: $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + ATP$.

The endoplasmic reticulum, or ER, exists in two forms. The rough endoplasmic reticulum is studded with ribosomes and is responsible for protein synthesis, while the smooth endoplasmic reticulum lacks ribosomes and is responsible for lipid synthesis and detoxification of harmful substances.

The Golgi apparatus is defined as an organelle that modifies, sorts, and packages proteins and lipids received from the endoplasmic reticulum before they are sent to their final destination, whether inside or outside the cell. It functions much like a shipping and processing center for the cell.

Ribosomes are the molecular machines responsible for protein synthesis. Protein synthesis refers to the two-stage process of transcription, in which messenger RNA is produced from a DNA template, and translation, in which ribosomes read the messenger RNA sequence to assemble a chain of amino acids into a functional protein.

Lysosomes are membrane-bound organelles containing digestive enzymes that break down waste materials, cellular debris, and foreign invaders. A defect in lysosomal enzymes can lead to a class of disorders known as lysosomal storage diseases.

3.4 Protein Synthesis in Detail

Transcription is defined as the process by which the enzyme RNA polymerase synthesizes a strand of messenger RNA, or mRNA, using one strand of DNA as a template. This occurs within the nucleus of eukaryotic cells.

Translation is defined as the process by which ribosomes decode the sequence of a messenger RNA molecule to produce a specific chain of amino acids, which then folds into a functional protein. Each group of three nucleotides in the mRNA, called a codon, corresponds to a specific amino acid according to the genetic code.

Many students find the distinction between transcription and translation to be one of the most difficult topics in introductory molecular biology, since both processes involve RNA but occur in different locations and produce different products.

3.5 Cell Division: Mitosis and Meiosis

Mitosis is defined as the process of nuclear division that produces two genetically identical daughter cells from a single parent cell, and is used for growth and tissue repair in multicellular organisms. Mitosis proceeds through the phases prophase, metaphase, anaphase, and telophase, commonly remembered with the mnemonic PMAT.

Meiosis is defined as a specialized type of cell division that produces four genetically distinct daughter cells, each with half the number of chromosomes as the parent cell, and is used specifically in the production of gametes for sexual reproduction.

A key exam tip is to remember that mitosis produces two diploid cells while meiosis produces four haploid cells; this distinction is one of the most frequently tested concepts on cell division exams.

3.6 Chapter Summary and Review Questions

In summary, the cell is the fundamental unit of life, composed of a plasma membrane enclosing a nucleus and a variety of specialized organelles including mitochondria, the endoplasmic reticulum, the Golgi apparatus, ribosomes, and lysosomes. Each organelle performs a distinct function that contributes to the overall survival and reproduction of the cell.

Students preparing for an exam on this chapter should be able to explain the structure and function of each major organelle, describe the process of protein synthesis from transcription through translation, and distinguish between mitosis and meiosis in terms of the number and genetic composition of resulting daughter cells.

Review question: What is the primary function of mitochondria in a eukaryotic cell? Review question: How does the structure of the rough endoplasmic reticulum relate to its function in protein synthesis? Review question: What are the key differences between mitosis and meiosis?